
School of Mathematics and Statistics

MT5864 Advanced Group Theory

Problem Sheet V: Soluble Groups

1. Let G and H be groups.
 - (a) Show that $(G \times H)' = G' \times H'$. Deduce that $(G \times H)^{(i)} = G^{(i)} \times H^{(i)}$ for all $i \geq 0$.
 - (b) Deduce that the direct product of two soluble groups is a soluble group. Could this result be deduced from other results proved in lectures?
2. Calculate the terms in the derived series for the following groups:

(i) S_3 ; (ii) D_8 ; (iii) D_{10} ; (iv) A_4 ; (v) S_5 .
3. Let p be a prime number and let G be a finite p -group. Show that G is soluble.
[Hint: Use the fact that $Z(G) \neq 1$ for a non-trivial p -group G .]
4. Show that a group of order 1352 is soluble.
[Hint: First find a non-trivial normal subgroup.]
5. (a) Let M and N be normal subgroups of a group G that are both soluble. Show that MN is soluble.
[Hint: $M \trianglelefteq MN$. What can you say about the quotient MN/M ?]
 (b) Deduce that a finite group G has a largest normal subgroup S which is soluble.
[‘Largest’ in the usual sense of containing all others. This normal subgroup S is called the *soluble radical* of G .]
 (c) Prove that S is the unique normal subgroup of G such that S is soluble and G/S has no non-trivial abelian normal subgroup.
6. (a) Let M and N be normal subgroups of a group G such that G/M and G/N are both soluble. Show that $G/(M \cap N)$ is soluble.
[Hint: Use the map $G \rightarrow G/M \times G/N$ given by $x \mapsto (Mx, Nx)$.]
 (b) Deduce that a finite group G has a smallest normal subgroup R such that the quotient G/R is soluble.
[‘Smallest’ in the usual sense of being contained in all others. This normal subgroup R is called the *soluble residual* of G .]
 (c) Prove that R is the unique normal subgroup of G such that G/R is soluble and $R' = R$.
[Hint: To show $R' = R$, remember that if H is a group with $N \trianglelefteq H$ such that H/N and N are soluble, then H is soluble.]

7. The following is an alternative way of proving that a minimal normal subgroup of a finite soluble group is an elementary abelian p -group.

Let G be a finite soluble group and M be a minimal normal subgroup of G .

- (a) By considering M' , prove that M is abelian. [Hint: $M' \text{ char } M$.]
- (b) By considering a Sylow p -subgroup of M , prove that M is a p -group for some prime p .
- (c) By considering the subgroup of M generated by all elements of order p , prove that M is an elementary abelian p -group.

8. Let $G = N \rtimes C$ be the semidirect product of a cyclic group $N = \langle x \rangle$ of order 35 by a cyclic group $C = \langle y \rangle$ of order 4 where the generator of C acts by inverting the generator of N : $y^{-1}xy = x^{-1}$.

Find a Hall π -subgroup H of G and its normalizer $N_G(H)$ when (i) $\pi = \{2, 5\}$, (ii) $\pi = \{2, 7\}$, (iii) $\pi = \{3, 5\}$ and (iv) $\pi = \{5, 7\}$.

9. Let p, q and r be distinct primes with $p < q < r$. Let G be a group of order pqr .

- (a) Show that G is soluble.
- (b) Show that G has a unique Hall $\{q, r\}$ -subgroup.
- (c) How many Hall $\{p, r\}$ -subgroups can G have? Can you construct examples to show that these numbers are indeed possible?

[Hints: Review Question 2 on Problem Sheet III. You may use the fact (that we quoted but did not prove) that any two Hall π -subgroups of a finite soluble group are conjugate. Semidirect products are good ways to construct groups with specified normal subgroups.]

10. A *maximal subgroup* of G is a proper subgroup M of G such that there is no subgroup H with $M < H < G$.

If p divides the order of the finite soluble group G , show that there is a maximal subgroup of G whose index is a power of p . [Hint: Consider Hall π -subgroups where $\pi = p'$ is the set of primes not equal to p .]

Show, by an example, that this is false for insoluble groups.

11. Let G be a finite soluble group whose order is divisible by k distinct prime numbers. Prove there is a prime p and a Hall p' -subgroup H such that $|G| \leq |H|^{k/(k-1)}$.

[Hint: Consider $|G| = p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$ and consider the smallest value of $p_i^{n_i}$.]

12. Let G be a group and H and K be subgroups of G .

- (a) Show that

$$|G : H \cap K| \leq |G : H| \cdot |G : K|.$$

[Show that there is well-defined map given by $(H \cap K)x \mapsto (Hx, Kx)$ into the set of pairs of cosets of H and of K .]

- (b) If $|G : H|$ and $|G : K|$ are coprime integers, show that

$$|G : H \cap K| = |G : H| \cdot |G : K|.$$

13. Let G be a finite soluble group and write

$$|G| = p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

where $p_1, p_2, \dots, p_k$ are the distinct prime factors of the order of G . Write p'_i for the set of all primes not equal to p_i and let Q_i be a Hall p'_i -subgroup of G for $i = 1, 2, \dots, k$. [The collection $\{Q_1, Q_2, \dots, Q_k\}$ is called a *Sylow system* for G .]

- (a) Show that $Q_1 \cap Q_2 \cdots \cap Q_t$ is a Hall $\{p_{t+1}, \dots, p_k\}$ -subgroup of G for each $t = 1, 2, \dots, k$. [Hint: Use induction and the previous question.]
- (b) Deduce that $P_r = \bigcap_{i \neq r} Q_i$ is a Sylow p_r -subgroup of G for $r = 1, 2, \dots, k$.
- (c) Show that $P_{k-1}P_k = P_kP_{k-1} = Q_1 \cap Q_2 \cap \cdots \cap Q_{k-2}$.
- (d) Deduce that $P_rP_s = P_sP_r$ for all $r, s \in \{1, 2, \dots, k\}$ with $r \neq s$.
[This shows that the family $\{P_1, P_2, \dots, P_k\}$ of Sylow subgroups, one for each prime divisor of $|G|$, is what is known as a *Sylow basis* for G .]